

20A Dedicated Circuit: DIY Safety Protocol

1. The 80% Rule

Continuous load (over 3 hours) should not exceed 80% of the breaker capacity. For a 20A breaker, this means a 16A maximum continuous draw.

2. Critical Wire Gauge Standards

Breaker	Wire Gauge (Copper)	Status
15 Amp	14 AWG	Standard
20 Amp	12 AWG	MANDATORY
30 Amp	10 AWG	MANDATORY

3. Voltage Drop Formula

$V_{drop} = (2 * L * R * I) / 1000$

Ensure V_drop is under 3% for sensitive hardware (AI Servers, Inverters).